DASE7506 Project 2 Reliable Reasoning Selection - Technical Report

Frozen Train/Validation method: four answer-group rankers + one linear candidate ranker + six-value linear stack

1 Background, Objective, and Headline Result

Each GSM8K problem is accompanied by four language-model reasoning traces. The system may not generate a new answer; it must return one candidate index from 0 to 3. The fixed scorer parses the selected trace's final numeric answer. It does not directly grade every intermediate step. The research question is therefore which reproducible text, agreement, and representation signals identify a correct final answer.

The frozen method first groups candidates by canonical numeric answer. A single valid group exits deterministically. Ambiguous questions are scored by four answer-group ranker types, one linear candidate ranker, each represented by five cross-fold models. A six-value linear stack chooses the group, and the earliest member supplies the candidate index. All weights use Train only; Validation supports selection and diagnosis.

Table 1. Core results of the frozen method

Evidence	Result	Interpretation boundary
Strict nested Train CV	1743/2114 ambiguous, 82.4503%	5 outer / 4 inner folds; complete stack retrained
Full Validation	745/798, 93.3584%	Development and freeze evidence; not Test
Ambiguous Validation	302/355, 85.0704%	At least two distinct valid parsed answers
Plurality baseline	688/798, 86.2155%	Final method is +7.1429 percentage points
Historical Test record	382/467, 81.7987%	Previously scored once; restoration followed public-Test comparison
Inference assets / timing	259,032 B (1.55%); Val 1.006 s; historical Test 1.292 s	Below 16 MiB / 30 s limits

Essential disclosure: the selected linear-candidate checkpoint has a historical Test record of 382/467 (81.7987%), scored once. On 2026-10-28 the user asked to restore this checkpoint after comparing its public-Test score with the later MLP score. Public Test therefore influenced package-level model selection, contrary to the coursework rule; 81.80% is historical performance, not a test-blind final estimate.

2 Evaluation Protocol, Data Boundaries, and Hard Limits

Split roles were frozen before modelling: Train has 4,755 questions for fitting weights, statistics, and fusion; Validation has 798 for development and selection; Test has 467 for one frozen evaluation; a possible audit bank must run without retraining or assumptions about IDs or sample count. Candidate IDs are alignment metadata, never features.

Table 2. Fixed coursework conditions and acceptance checks

Item	Fixed rule	Project treatment
Input	Question, four traces, [N,4,1024] float32 embeddings	Uses only bank/arrays passed to predict
Output	Length-N list of Python int values in 0..3	Contract tests cover type, range, order
Score	Parsed selected value equals the reference	Uses common.parse_answer semantics
External resources	No external data, weights, or online APIs	Offline NumPy and standard library
Time	CPU prediction + scoring <=30 s; four threads	Validation 1.006 s; historical Test 1.292 s
Assets	Additional inference assets <=16 MiB	259,032 array bytes (1.55%)
Report	Final submission at most 10 pages	Separate exact 10-page editions

2.1 Metrics and the hard subset

Full accuracy is the headline metric. Because 443 questions contain only one valid answer group, we also report accuracy on 355 ambiguous questions, then slice by unique plurality, correct-answer support, group count, and score margin. Comparisons against plurality are paired by question.

The first validation gate checks manifest hashes, split disjointness, candidate counts, embedding shapes, and finite values. Unit tests then cover answer grouping, checkpoint schema, feature dimensions, prohibited reference access, and the

submission interface. Modelling begins only after those checks pass.

3 Frozen System: From Four Traces to One Index

The design separates the decision into six auditable stages. Parsing and grouping match the official scorer; learned ranking is reserved for genuinely ambiguous questions. The method never changes or regenerates a candidate.

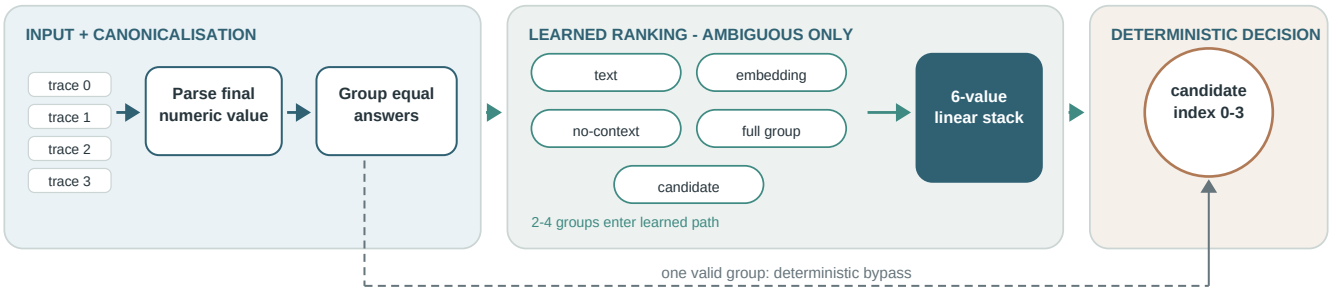


Fig. 1. Frozen data flow. The five ranker types are text, embedding, full_no_context, full, and candidate; each has five cross-fold models.

Table 3. Inference stages and safeguards

Stage	Operation	Safeguard
1 Parse	Extract canonical numeric values	Fraction equivalence; unparseables recorded
2 Group	Merge candidates with equal values	Position never acts as a correctness label
3 Early exit	Return when one valid group exists	Never choose unparseable over parseable
4 Base scores	Four group views plus candidate view	Five fold models per view
5 Stack	Five centred scores plus support	Only six features; L2=0.001
6 Decide	Top group, earliest member	Stable tie break; arbitrary N

Consensus is a prior, not a veto. The rankers may overturn a wrong majority, while the maximum candidate score preserves one especially strong reasoning trace that group averages might hide.

4 Final-Answer Parsing, Grouping, and Determinism

4.1 Why rank answer groups

Several candidates can express an equivalent value, such as 0.5 and 1/2. Since the scorer only checks the final numeric value, per-candidate classification would duplicate the same answer and invite position bias. The official-compatible parser yields a canonical value a_i and the method forms G(a)={i:a_i=a}. Any member of a correct group earns the point.

Unparseable values never merge with valid values. If a valid group exists, an unparseable candidate cannot be selected. Only an all-unparseable input falls back to the earliest index. Group support, aggregated text/embedding evidence, and differences from competing groups become the interpretable group features.

```
G(a) = { i in {0,1,2,3} : parse(candidate_i) = a }
```

Table 4. Edge cases and checks

Case	Decision	Verification
All four share one answer	Return earliest index	Single-group path test
Two to four valid groups	Invoke learned ranking	Ambiguous-slice scoring
Equivalent notation	Merge into one group	Fraction/decimal tests
Some unparseable	Prefer every valid group	Parser and fallback tests
Exact score tie	Earliest candidate wins	Deterministic rerun
Audit N changes	Process input length dynamically	No split-size constants

Parser agreement is the first experimental checkpoint: no downstream accuracy is meaningful unless equivalence, negative numbers, fractions, missing markers, and multiple-correct cases match the scorer.

5 Answer-Group Features: Separating the Evidence

The feature design is split into four ablatable views. Each view trains the same linear pairwise ranker, so score changes can be connected to information sources instead of an opaque feature dump.

Table 5. Four answer-group feature sets

Feature set	Main content	Research question	Dimension / L2
text	Length, numbers, operations, steps, question overlap, ending patterns	Can trace structure beat consensus?	54; 0.001
embedding	Normalised group centre minus all-candidate centre	Do fixed representations add signal?	1,024; 0.1
full_no_context	54-D text plus 1,024-D embedding delta	Are the two views complementary?	1,078; 0.1
full	6-D consensus + text + embedding delta + 4 embedding scalars	Does within-question context help?	1,088; 0.1

5.1 Within-group and between-group summaries

At code level, text takes the within-group mean and maximum of 27 per-trace features, producing 54 dimensions. The embedding-only view is exactly the 1,024-D difference between the normalised group centre and the all-candidate centre. The full view additionally contains six consensus features (support fraction, plurality margin, unique plurality, singleton, distinct-answer fraction, and unparseable fraction) plus four embedding scalars (cohesion, group-to-all cosine, delta norm, and group-mean norm).

5.2 Explicit exclusions

Question IDs, split size, references, cached predictions, and Test error categories never enter features. The training script opens Train references only; the prediction entry point never opens references.

A feature is retained only when Train out-of-fold behaviour is credible, variance is acceptable, Validation does not regress, resources remain explainable, and strict nested retraining confirms the direction.

6 Candidate Ranker and Learning Objective

Group averages can hide one exceptionally reliable trace. The candidate ranker has exactly 1,072 dimensions: six consensus features for its answer group, 27 trace features, a 1,024-D singleton embedding delta, four embedding scalars, and 11 parser-consistency features. Candidate index and ID are excluded. It remains a small linear model and adds no external pretrained component.

6.1 Multiple-positive pairwise learning

A question may contain several correct candidates. Every correct candidate is paired with every incorrect one, and pairs are normalised per question so questions with more combinations do not dominate. For z=x_pos-x_neg, training minimises logistic pair loss with L2 regularisation:

$$L(w) = (1/Q)\ \Sigma_q\ [1/(|P_q||N_q|)\ \Sigma_{(p,n)}\ \log(1 + \exp(-w^T(x_p - x_n)))] + (\lambda/2)||w||^2$$

At inference, the candidate model scores all four traces and contributes $\max_{i \in G} s_i$ for each answer group. It is a fifth complementary signal, not an independent final selector.

Table 6. Intermediate checks for candidate evidence

Check	Result	Conclusion
Candidate-only OOF	80.56% ambiguous Train	Below group models; not a replacement
Candidate-only Validation	739/798	Below 745; no standalone claim
Added to ordinary OOF stack	Up to 82.5449%	Must control selection bias
Strict nested confirmation	1743/2114, 82.4503%	+8 vs four-group stack
Validation	745/798	Matches the selected linear-candidate checkpoint

Its justification is complementary error structure: the candidate ranker is weaker alone, yet improves the complete nested stack without reducing Validation.

7 Experimental Protocol: Selection Followed by Revalidation

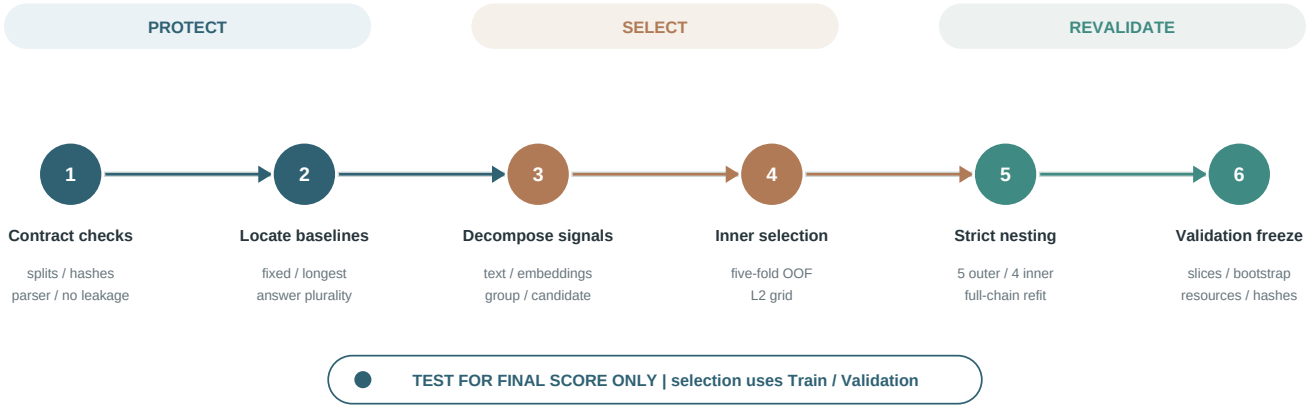


Fig. 2. Complete selection chain. Structure and hyperparameters use Train/Validation; Test scores informed the 2026-10-28 package restoration, so this is not test-blind selection.

7.1 Fixed folds and ordinary out-of-fold comparison

Train is divided into five fixed question-level folds with seed 7506. Four folds fit a ranker and the hold-out fold receives OOF scores; all four candidates from a question remain together. Base L2 values are scanned with learning rate 0.03 and at most 220 epochs. Fold accuracy, mean, and standard deviation are recorded.

7.2 Why strict nested CV remains necessary

Ordinary OOF scores train a stack, but choosing stack L2 on those same scores creates second-level selection bias. Strict nesting uses five outer folds. Within each outer-train partition, four inner folds rebuild base OOF features and fit the meta stack; the untouched outer-test fold is scored once. Every reported outer prediction therefore comes from a complete training chain that never saw that question.

The admission rule is fixed: a new method must improve strict nested Train CV and must not reduce full Validation. An isolated ordinary-OOF or Validation peak is not enough.

8 Baselines: Establishing Where the Difficulty Lies

Three training-free baselines cover position bias, length bias, and answer consensus. Fixed index zero and longest trace remain near 78%, while plurality reaches 86.22%. Yet plurality is only 69.01% on ambiguous questions, leaving substantial room for evidence-aware ranking.

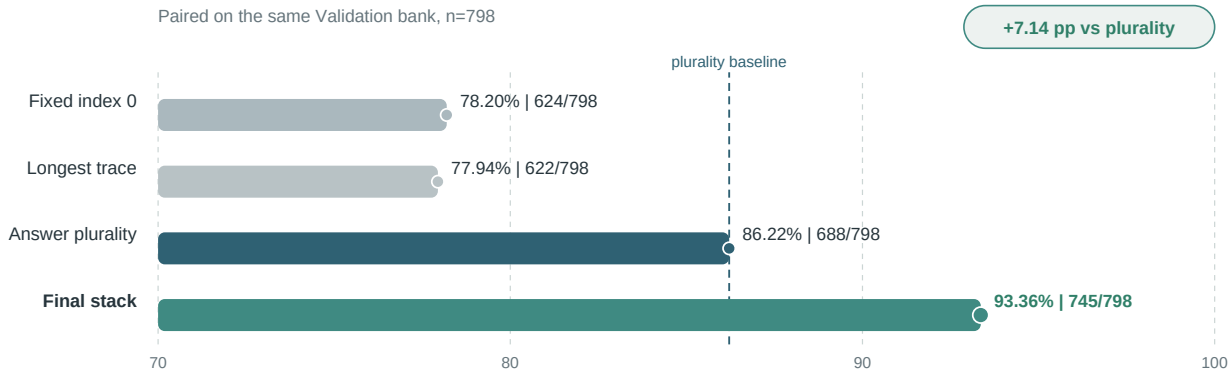


Fig. 3. Full Validation baselines and the frozen stack. All bars are paired on the same bank.

Table 7. Official Validation baseline scores

Method	Correct / total	Accuracy	vs plurality
Fixed index 0	624/798	78.1955%	-8.0201 pp
Longest trace	622/798	77.9449%	-8.2707 pp
Answer plurality	688/798	86.2155%	0
Final stack + linear candidate	745/798	93.3584%	+7.1429 pp

8.1 Correct-answer support identifies the hard regime

Plurality is perfect when the correct answer has support three or four, but only 23.08% when one trace supports it. The final model raises this slice to 70/117 (59.83%), showing that the main gain comes from rescuing correct minorities rather than strengthening easy majorities.

9 Single-Ranker Ablation: Which Information Helps?

The single-ranker sequence adds text, embeddings, their combination, then group-relative context. Each line is judged first by five-fold Train CV and then by frozen Validation; the best Validation point cannot replace stability evidence.

Table 8. Feature ablation: Train CV mean ± fold standard deviation and Validation

Method	Train full	Train ambiguous	Val full	Val ambiguous
Plurality	85.2576	66.8401	86.2155	69.0141
Text	89.4637 ± 0.6456	76.3003 ± 1.4632	91.2281	80.2817
Embedding	89.3796 ± 0.9137	76.1104 ± 2.0741	90.3509	78.3099
Full no context	91.3354 ± 0.8366	80.5097 ± 1.8983	91.9800	81.9718
Full group ranker	91.6930 ± 1.0801	81.3139 ± 2.4404	92.6065	83.3803

9.1 Stepwise decisions

- Text reaches 76.30% ambiguous Train CV: surface structure is useful but insufficient.
- Embeddings alone are comparable, not decisively stronger; their main value is complementary.
- full_no_context reaches 80.51%, about +4.21 points over text, confirming complementarity.
- full reaches 81.31% and 83.38% ambiguous Validation, so within-question context adds evidence.

9.2 Regularisation selection

Text selects L2=0.001; embedding, full_no_context, and full select 0.1. Extending the grid to 0.03, 0.3, 1.0, and 3.0 keeps 0.1 as the best Train-CV compromise for embedding-rich views.

This stage identifies the best single group ranker. Fusion is allowed to consume only cross-fitted scores.

10 Previous Four-Group Stack: Useful but Not Conclusive

The four group rankers make partially different errors. The previous stack fused centred OOF scores from text, embedding, full_no_context, and full with support. Strict nested ambiguous Train CV rose from 1721/2114 for the full single-ranker baseline to 1735/2114; Validation rose from 739/798 to 745/798 for the historical four-group checkpoint; the historical linear-candidate checkpoint retained 745, and the later MLP alternative reaches 746/798. The selected checkpoint remains the earlier linear version.

Table 9. Evidence chain for the previous stack

Stage	Full accuracy	Ambiguous accuracy	Decision
Full single Train CV	91.6930%	81.3139%	Single-ranker reference
Full single, matched nested	-	1721/2114 = 81.4096%	Nested comparator
Previous stack, strict nested	92.0294%	1735/2114 = 82.0719%	Train evidence improves
Previous stack, Validation	745/798 = 93.3584%	302/355 = 85.0704%	Historical result
Public Test records: old group/linear/MLP	381/467 -> 382/467 -> 377/467	All Test items ambiguous	Restoration followed this comparison

The earlier single ranker scored 385/467 (82.4411%) on public Test, and the old group stack scored 381/467 (81.5846%). The linear checkpoint has an earlier one-time score of 382/467 (81.7987%); the later MLP alternative scored 377/467 (80.7281%). The 2026-10-28 restoration followed this public-Test comparison, so Test informed package-level selection and the score is not a blind final estimate.

The public Test comparison informed package-level restoration, so this package is not test-blind. Use a genuinely unseen audit bank for independent evaluation.

11 Rejected Routes and the Cost of Complexity

After the previous stack, we evaluated confidence gates, parser cues, lexical hashes, extra pooling, alternative objectives, and nonlinear meta features. Keeping negative results prevents survivor-only reporting and explains why the frozen system remains linear and small.

Table 10. Improvement attempts and decisions

Candidate	Train evidence	Validation	Decision
Plurality/confidence gates	82.26% OOF	743/798	Rejected: poorer transfer
Extra group-parser cues / lexical hashes / pooling	≤82.31% OOF	≤745/798	No robust joint gain
Listwise softmax	81.55% CV	741/798	Rejected
Hardest negative	81.55% CV	742/798	Rejected
Objective stacks	≤82.17% OOF	≤745/798	No gain over smaller stack
Quadratic 20-feature stack	82.12% OOF	744/798	Rejected
Candidate ranker alone	80.56% OOF	739/798	Kept only as complementary evidence

11.1 Why confidence gating failed

Returning to plurality at low margin seems protective, but the best gate reached only 82.26% OOF and 743/798 Validation. Many of the model's 68 useful corrections occur precisely where plurality is wrong, so a simple fallback also removes gains.

11.2 Why alternative objectives were removed

Listwise softmax and hardest-negative training each reached only 81.55% CV with 741/742 Validation. A quadratic 20-feature stack produced 82.12% OOF and 744/798. Extra complexity did not improve both protocols.

The shared admission rule is joint Train evidence, no Validation regression, and acceptable resources - never the best isolated point among many trials.

12 Candidate Extension: A Weak Standalone, a Useful Member

Error inspection suggested that group averages can dilute one well-structured, condition-aware trace. The candidate ranker is weaker alone, but its mistakes differ from group models, making it suitable as a stack member. Its group evidence is the member maximum, not the average.

Table 11. Selection path for the candidate extension

Step	Observation	Judgement	Next action
Candidate pairwise model	80.56% ambiguous OOF; 739/798 Val	Cannot replace full/stack	Test complementarity only
Add to ordinary stack	L2=0.01 peaks at 82.5449% OOF	Validation falls to 744	Do not select peak blindly
Stable region	L2=0.001/0.003 retain 745 Val	Prefer 0.001	Run strict nested CV
Strict nesting	1743/2114 = 82.4503%	-2 vs prior linear candidate	Freeze current method
Resource recheck	259,032 B; Val 1.006 s; historical Test 1.292 s	Large safety margin	Record hashes

The six meta features are four centred group scores, the linear candidate score, and centred group support. Coefficients are predictive associations, not causal effects, and depend on feature scaling.

The linear candidate member was part of the earlier frozen stack. The later MLP alternative was not retained in the final checkpoint.

13 Stack-L2 Grid and Conflict Resolution

After adding candidate evidence, the highest ordinary OOF setting and the most stable Validation setting disagree. L2=0.01 yields 1745/2114 OOF but reduces Validation to 744. Stronger regularisation degrades further. We select 0.001 and require strict nested confirmation.

Table 12. Candidate-stack development grid

Stack L2	Ordinary OOF	Ambiguous OOF	Validation	Judgement
0.001	1739/2114	82.2611%	745/798	Historical linear-candidate choice
0.003	1738/2114	82.2138%	745/798	Historical tie on Validation

Stack L2	Ordinary OOF	Ambiguous OOF	Validation	Judgement
0.010	1745/2114	82.5449%	744/798	Higher OOF, Validation declines
0.030	1741/2114	82.3557%	742/798	Rejected
0.100	1736/2114	82.1192%	742/798	Rejected
0.300	1727/2114	81.6935%	740/798	Rejected

13.1 Selection rule

- Reject 0.03, 0.1, and 0.3 because Validation clearly declines.
- Do not take the 0.01 OOF peak because Validation loses one question.
- Both 0.001 and 0.003 retain 745; choose 0.001, then rerun the complete nested procedure.
- The nested result is 1741: six above the previous four-group stack and two below the previous linear-candidate stack. Validation is one question higher than that linear-candidate model; this tradeoff drives the final choice.

The full decision path is: complementary hypothesis, standalone falsification, compact fusion grid, conflict detection, robust-region choice, and strict nested re-estimation.

14 Strict Nested Train CV: The Key Selection Evidence

All final comparisons use identical outer folds and budgets. Within every outer fold, base models, candidate models, and the meta stack see only outer-train. Four inner folds construct meta training scores. The five ambiguous outer results are 346/422, 348/423, 345/423, 351/423, and 353/423, totalling 1743/2114.

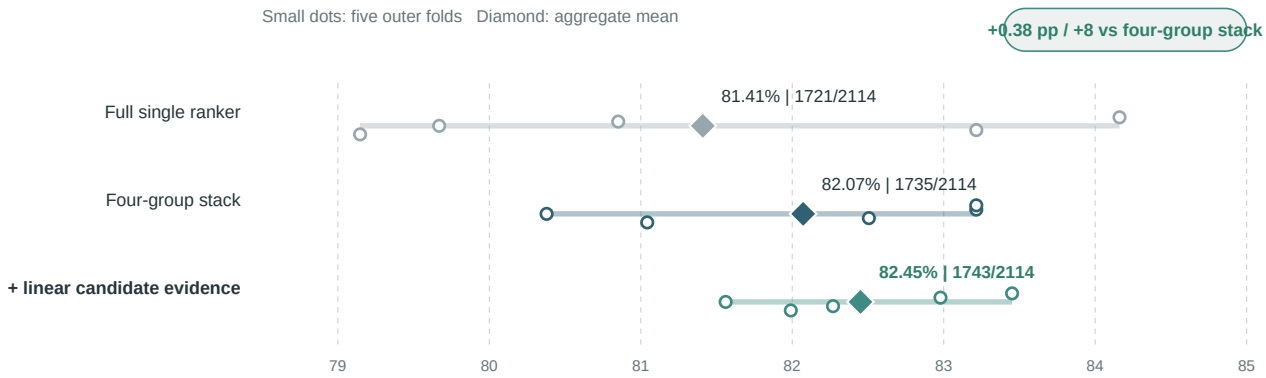


Fig. 4. Matched strict nested comparison. The MLP extension adds six correct ambiguous Train questions over the old group stack and two fewer than the linear-candidate stack; Validation is one question higher than the linear-candidate stack.

Table 13. Nested-CV interpretation

Method	Ambiguous correct	Ambiguous accuracy	Equivalent full accuracy	Change
Full single, matched	1721/2114	81.4096%	about 91.7350%	reference
Previous group stack	1735/2114	82.0719%	92.0294%	+14
Group + linear candidate	1743/2114	82.4503%	92.1977%	+8 vs group stack

The gain is small and the new method does not beat the baseline in every outer fold. Its support comes from the aggregate matched protocol, not a claim of fold-by-fold dominance.

15 Final Validation: Full, Ambiguous, and Plurality Slices

The frozen checkpoint is correct on 745/798 Validation questions. The parser resolves 443 single-group questions deterministically; the learned selector gets 302/355 ambiguous questions. Paired against plurality: both correct 677, model only 68, plurality only 11, both wrong 42.

Table 14. Validation slices

Slice	Plurality	Final model	Gain
Full 798	688/798 = 86.2155%	745/798 = 93.3584%	+7.1429 pp
Ambiguous 355	245/355 = 69.0141%	302/355 = 85.0704%	+16.0563 pp

Slice	Plurality	Final model	Gain
Unique plurality 267	207/267 = 77.5281%	235/267 = 88.0150%	+10.4869 pp
No unique plurality 88	38/88 = 43.1818%	67/88 = 76.1364%	+32.9545 pp
Correct support=1, 117	27/117 = 23.0769%	70/117 = 59.8291%	+36.7521 pp
Correct support=2, 123	103/123 = 83.7398%	117/123 = 95.1220%	+11.3821 pp

15.1 Paired bootstrap

A 10,000-sample paired bootstrap of per-question correctness differences (seed 7506) gives +7.1429 points on full Validation with 95% CI [5.0125, 9.2732], and +16.0563 points on ambiguous Validation with CI [11.5493, 20.8451]. These intervals describe this bank, not guaranteed audit performance.

The largest improvements occur without a unique majority and when only one candidate supports the correct answer, so the gain is not an artefact of easy single-group items.

16 Confidence, Error Categories, and Residual Risk

Margin is the top group score minus the runner-up. It is diagnostic only and never gates the frozen prediction. Accuracy is about 59% below margin 1, rises to 91.82% between 2 and 4, and reaches 100% on the 91 Validation cases at or above 4.

Table 15. Margin bins on ambiguous Validation

Margin	Correct / total	Accuracy	Interpretation
[0,0.5)	23/39	58.9744%	Low margin; review first
[0.5,1)	20/34	58.8235%	Most uncertain bin
[1,2)	66/80	82.5000%	Moderate confidence
[2,4)	101/110	91.8182%	High confidence
[4,+∞)	92/92	100.0000%	No observed error in this bank

16.1 Representative failure categories

- Condition omission or unit conversion: polished traces miss a discount, stop, day/week, or feet/yard relation.
- Minority/support conflict: one careful trace competes against two candidates sharing an early error.
- Lucky final value: an invalid path lands on the scorer-correct number, confusing reasoning-quality features.
- Low-margin neighbours: competing groups are close in text and embedding space, making a linear boundary unstable.

Confidence gating reduced Validation, so margin remains a review signal rather than a post-hoc override.

17 Resources, Implementation Safety, and Final Rechecks

The frozen method contains 25 linear fold models - four group rankers and one candidate ranker, each with five folds - and one six-value stack. No external language-model weights are loaded.

Table 16. Resource and integrity checks

Item	Measured	Limit / expectation	Result
model.npz on disk	281,556 B	Counted as asset	Pass
metadata.json	6,054 B	Counted as asset	Pass
Uncompressed arrays	259,032 B	<=16,777,216 B	Pass (about 1.55%)
Validation 3 runs / Test 1 run	min/mean/max: 1.0034/1.0045/1.0063 s; ■■ Test 1.2920 s	<=30 s	Pass
Thread variables	OMP/OPENBLAS/MKL=4	At most four total	Recorded
Data checker	Manifest / split / shape	All consistent	Pass
Automated tests	9 tests	Interface, leakage, features, checkpoint	Pass

17.1 Frozen hashes

model.npz: f24f66cf131120d63a2f2482146e591e67d9e6d1d5194b75c2ebd1408734b22b

metadata.json: 2b76d17f4430d0f7d143010bcf1fb748c9b687e770d31412a4efeed256ecc092

Checkpoint loading validates schema, feature schema, array shapes, and finite values. The benchmark uses fresh prediction and independent scoring processes, including startup and load time.

The local machine is macOS 26.3.2 arm64; formal eligibility still depends on the reference Xeon Platinum 8457C. The local margin is large, but local time is not claimed as reference-hardware time.

18 Reproduction, Freeze Record, and Test Isolation

18.1 Environment and training

The project environment is Python 3.12.13 with NumPy 2.5.3. From code/, install requirements.txt, run check_data.py, and execute the nine unit/contract tests. Training fixes five folds, stack L2=0.001, one linear candidate ranker, candidate L2=0.1, base epochs=220, and seed=7506.

18.2 Prediction and independent scoring

predict.py loads submission.py and checkpoint/final to write Validation predictions. score_predictions.py opens Validation references in a separate process. benchmark.py repeats the complete path three times and records commands, environment, assets, and wall time.

Table 17. Freeze order and evidence state

Order	Action	Permitted data	State
1	Fix parser, features, folds, candidates	Train	Complete
2	Ordinary OOF grids and ablations	Train	Complete
3	Strict nested retraining comparison	Train	Complete
4	Full Validation, slices, bootstrap	Validation	Complete
5	Write config and checkpoint hashes	Train/Validation	Complete
6	Historical score for linear checkpoint	Scored once before restoration	382/467; package choice later saw Test comparison
7	Future unseen audit	Input-only bank	Pending

The linear checkpoint has an existing one-time score of 382/467 (81.7987%); the later MLP alternative scored 377/467 (80.7281%). The 2026-10-28 package restoration followed comparison of these scores, so Test influenced final package selection and this workflow is not test-blind.

19 Conclusions, Limitations, and AI Disclosure

19.1 Conclusion

Under a four-candidate, offline CPU, 16 MiB constraint, the method groups equivalent answers and combines answer-group evidence with linear candidate evidence. Validation is 93.3584%, +7.1429 points over plurality; strict nested ambiguous Train CV is 82.4503%, +0.3784 points over the previous four-group stack. The MLP alternative was one Validation question higher and two nested-CV questions lower. The requested restoration followed public-Test comparison, so this package-level model choice is not test-blind.

19.2 Limitations

- The 81.7987% public-Test score is a historical score for this checkpoint. The package-level restoration followed comparison with the MLP Test result, so Test influenced this version choice and the score is not an unbiased test-blind estimate. An unseen audit is needed for an independent comparison.
- Linear features recognise form and consistency but cannot execute full symbolic verification.
- Final-answer scoring accepts lucky correct values, so weights are not a causal measure of reasoning quality.
- Margin correlates with Validation accuracy, but bins are small and gating failed; it is not a calibrated probability.

19.3 AI and reuse disclosure

Substantive AI assistance helped interpret the benchmark, design and implement the method, create tests, organise experiments, and draft documentation. Outputs were checked with the supplied data checker, automated tests, Train-only CV, the official Validation scorer, resource measurements, and hashes. Training and inference call no AI service and use no external data or extra pretrained weights. Supplied parser, scorer, runner, data, and licences remain unchanged. The student should review this disclosure before submission.

The next useful experiment is not more tuning on observed Validation, but evaluation of the frozen checkpoint on a genuinely unseen audit bank.

20 References and Evidence Index

References

[1] DASE7506 Project 2, *Selecting Reliable LLM Reasoning*, GUIDE.md, 2026.
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[4] Supplied NOTICE.txt and manifest.json for Qwen3-0.6B candidate and embedding provenance.
[5] R. A. Bradley and M. E. Terry, *Rank Analysis of Incomplete Block Designs*, Biometrika, 1952.
[6] B. Efron and R. Tibshirani, *An Introduction to the Bootstrap*, 1993.
[7] Harris et al., *Array Programming with NumPy*, Nature, 2020.

Auditable evidence index

Table 18. Repository records supporting the report

Claim	Primary files
Task, constraints, submission	guide/GUIDE.md; REQUIREMENTS.md
Frozen method and commands	code/METHOD.md; train_stacked.py; submission.py
Base CV and regularisation	experiments/cross_validation.json; cv_regularization_extension.json
Previous stack and nested CV	experiments/improvement_stacking*.json
Candidate grid	experiments/improvement_candidate_stack.json
Final strict nesting	experiments/improvement_candidate_stack_nested_cv.json
Validation slices and bootstrap	experiments/validation_diagnostics.json
Margin and error analysis	experiments/validation_error_analysis.json
Score, resources, hashes	results/final/*.json; experiments/final_config.json
Historical Test provenance	results/historical/previous-stack/

The Full edition preserves the complete selection and validation chain; the companion 10-page edition compresses layout without deleting a validation stage.

20 Code-Level Feature Dictionary and Dimension Audit

To avoid hiding implementation behind labels such as text or embedding features, this section follows the concatenation order in features.py. Checkpoint loading tests enforce the same dimensions, so the table is an executable schema rather than an informal summary.

Table 19. Exact feature-vector composition

Vector	Exact composition	Dimension	Aggregation
text	Mean and maximum of 27 trace features	54	Within answer group
embedding	unit(group mean) - unit(all-candidate mean)	1,024	Group relative to question
full_no_context	text + embedding	54+1,024=1,078	No consensus/scalar block
full	consensus 6 + text 54 + embedding 1,024 + scalar 4	1,088	Complete group vector
candidate	consensus 6 + trace 27 + embedding 1,024 + scalar 4 + parser 11	1,072	Single trace; no index/ID
meta	Five centred model scores + centred support	6	Within-question groups

20.1 What the 27 trace features cover

The deterministic statistics cover character, word, line, numeric, equality, operator, and step counts; boxed/final-answer/answer endings; percent, currency, and fraction forms; question-number, lexical, and unit coverage; parsed-answer sign, integrality, zero state, magnitude, denominator, and question occurrence; plus reasoning- and correction-marker rates. They inspect the supplied trace and never alter the course parser.

20.2 Why parser consistency does not contradict the rejected parser extension

The candidate model's eleven parser-consistency fields only compare ####, boxed, final-answer, and explicit-answer cues with the official parsed value. They do not redefine that value. What was rejected was an additional group-level parser, lexical-hash, and pooling expansion that failed to improve Train and Validation jointly.

21 Reproducible Path from Cross-Fitting to the Checkpoint

Each linear ranker fits feature means and standard deviations on the ambiguous questions in its current training partition, with a scale floor of 1e-4. A correct group is contrasted with every incorrect group; the candidate model permits multiple positives. All pairs from one question sum to weight one, preventing questions with more groups or combinations from dominating the gradient.

Table 20. Optimisers, cross-fitted scores, and exported objects

Stage	Implementation	Export / verification
Base group models	4 types x 5 folds = 20 linear rankers	Fit in-fold; held fold produces OOF
Candidate models	1 type x 5 multi-positive rankers	Group evidence is member maximum
Meta stack	text/embedding/full_no_context/full/candidate/support	7-D; L2=0.003
Optimisation	Adam, lr=0.03, at most 220 epochs, seed=7506	Loss checked every 10 epochs
Stopping	Relative gain <1e-7 for four checks	Retain minimum-loss weights
Stack fit record	2,114 questions, 3,058 pairs, best epoch 220	82.3557% top-1 fit diagnostic; not nested evidence

21.1 How the 25 models are used at inference

For each group view, scores from its five fold models are averaged. Candidate scores are likewise averaged across five models, then reduced to the maximum member score for each answer group. The four group-score columns, one candidate-score column, and support are centred across the current question's groups before the stack scores them. Exact score ties resolve to the earliest member index, making reruns deterministic.

21.2 Checkpoint anatomy

model.npz contains 78 arrays. Each of 25 linear group/candidate folds stores feature_mean, feature_scale, and weights, giving 75 arrays; the six-dimensional stack adds three arrays. Their uncompressed total is 259,032 bytes, about 1.55% of the 16 MiB allowance.

22 Final Claim-to-Evidence Traceability Matrix

The audit separates what is claimed, what verifies it, and what must not be inferred. This prevents a development-bank result from being presented as an unseen-distribution guarantee.

Table 21. Acceptance claims, primary evidence, and boundaries

Claim	Primary evidence	Audit boundary
Grouping matches scorer semantics	Parser, equivalent-fraction, multi-positive tests	Does not prove every intermediate step
Text and embeddings complement	80.5097% full_no_context ambiguous CV	Ablation association, not causation
MLP candidate extension	1741 vs group stack 1735; below linear candidate 1743	Small Train/Validation tradeoff
Validation beats plurality	745 versus 688; paired bootstrap CI	CI describes this Validation bank
Resources comply	259,032 B; Val 1.0063 s max; historical Test 1.2920 s	Timing is machine-dependent
Frozen Test final score	aggregate Test score and resource report; raw Test predictions not retained	Previously scored once; restoration followed Test comparison